

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Project Title:

Telemedicine Slot Booking & Prescription Portal

Problem Statement #11 – Healthcare & Telemedicine
SRN-PES1UG24CS439

1. INTRODUCTION

The Telemedicine Slot Booking & Prescription Portal is a secure healthcare consultation system that allows patients to view doctors and their medical specialties, book video consultation slots, access secure consultation rooms, and download digitally signed prescriptions.

2. PURPOSE

The purpose of this system is to provide a secure and convenient platform for patients to select suitable physicians, book available video consultation slots, attend remote consultations, and obtain digitally signed prescriptions.

Physicians can conduct consultations and generate and digitally sign prescriptions.

3. SCOPE

The system includes the following functions:

- View doctors and their medical specialties.
- View available video consultation slots.
- Book a video consultation slot.
- Make payment for the consultation.
- Access a secure consultation room.
- Conduct video consultation.
- Generate prescriptions.
- Digitally sign prescriptions.
- Download digitally signed prescriptions.
- Protect patient and consultation data.
- Provide fast loading of available consultation slots.

4. SYSTEM ACTORS

4.1 Patient

The patient can:

- View doctor specialties.
- View available doctors.
- Book a consultation slot.
- Make payment.
- Access the video consultation.
- Download the prescription.

4.2 Physician

The physician can:

- View consultation information.
- Conduct consultations.
- Generate prescriptions.

- Digitally sign prescriptions.

4.3 Payment Gateway

The payment gateway processes the payment required for consultation booking.

5. FUNCTIONAL REQUIREMENTS

FR-001 – Digital Prescription

Description:

The system shall enable physicians to author and digitally sign structured prescriptions that are automatically compiled into downloadable PDF records.

Priority: High

Acceptance Criteria:

Pass – A valid cryptographic signature is attached to the prescription PDF.

Fail – A prescription is generated without doctor license validation.

Rationale:

Ensures that prescriptions are authentic and digitally signed.

FR-002 – View Doctors and Specialties

Description:

The system shall allow patients to view doctors and their medical specialties.

Priority: High

Acceptance Criteria:

Pass – Selecting a specialty displays doctors belonging to that specialty.

Fail – Doctors from unrelated specialties are displayed.

Rationale:

Helps patients select an appropriate physician.

FR-003 – Book Video Consultation Slot

Description:

The system shall allow patients to view and book available video consultation slots with a selected physician.

Priority: High

Acceptance Criteria:

Pass – The selected slot is booked and associated with the patient.

Fail – The system allows two patients to book the same slot.

Rationale:

Provides the core appointment-booking functionality.

FR-004 – Secure Consultation Room

Description:

The system shall provide patients with a secure consultation room link after successful appointment booking.

Priority: High

Acceptance Criteria:

Pass – A unique consultation link is available to the booked patient.

Fail – An unbooked patient can access the consultation room.

Rationale:

Enables secure remote consultation.

FR-005 – Download Prescription

Description:

The system shall allow patients to download their digitally signed prescriptions after a consultation.

Priority: Medium

Acceptance Criteria:

Pass – The patient can download the signed prescription as a PDF.

Fail – The patient can download an unsigned prescription.

Rationale:

Allows patients to access their prescription records.

6. NON-FUNCTIONAL REQUIREMENTS

NFR-001 – Security

Description:

All video session metadata and patient clinical notes must be encrypted in transit using TLS 1.3 and at rest with AES-256.

Priority: High

Acceptance Criteria:

Pass – Security testing confirms TLS 1.3 and AES-256 encryption.

Fail – Required encryption is not applied.

Rationale:

Protects sensitive patient healthcare information.

NFR-002 – Performance

Description:

The system shall load available consultation slots within 3 seconds under normal operating conditions.

Priority: High

Acceptance Criteria:

Pass – At least 95% of slot-search requests load within 3 seconds.

Fail – More than 5% of requests take longer than 3 seconds.

Rationale:

Ensures a responsive booking experience.

7. SYSTEM WORKFLOW

1. Patient views doctor specialties.
2. Patient views available doctors.
3. Patient selects a physician.
4. Patient views available consultation slots.
5. Patient selects a consultation slot.
6. Patient makes the required payment.
7. The system confirms the appointment.
8. The patient receives access to the secure consultation room.
9. Physician conducts the consultation.
10. Physician generates the prescription.
11. Physician digitally signs the prescription.
12. Patient downloads the digitally signed prescription.

8. USE CASES

The main use cases of the system are:

- View Doctor Specialties
- View Available Doctors
- Book Consultation Slot
- Make Payment
- Access Video Consultation
- Reschedule Consultation
- Conduct Consultation
- Generate Prescription
- Digitally Sign Prescription
- Download Prescription

9. SYSTEM SECURITY

The system shall protect sensitive healthcare information.

Security includes:

- Secure consultation-room access.
- Validation of patient access.
- Encryption of patient clinical notes.
- Encryption of video session metadata.
- Digital signing of prescriptions.
- Prevention of unauthorized consultation-room access.

10. SYSTEM PERFORMANCE

The system should provide quick access to available consultation slots.

The target is that at least 95% of slot-search requests should load within 3 seconds under normal operating conditions.

11. CONCLUSION

The Telemedicine Slot Booking & Prescription Portal provides a secure platform for patients and physicians to conduct remote healthcare consultations. The system supports doctor discovery, appointment booking, secure consultation access, payment, prescription generation, digital signing, and prescription download while maintaining security and performance requirements.